Topological Integrity Gravity III:

Conditional Vacuum Admissibility and Bounded Curvature Structure

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May 17, 2026

Abstract

We develop a conditional operator-theoretic perturbative framework for a representative bounded-curvature vacuum sector compatible with the cubic branch structure established in Topological Integrity Gravity II (TIG2).

The framework is based on a structurally regulated spherically symmetric geometry preserving Schwarzschild-compatible asymptotics while introducing a finite representative core scale. We analyze the resulting perturbative radial operator structure and formulate conditions for quadratic-form control, self-adjoint realization, spectral admissibility, and bounded linear perturbative evolution.

The present work is formulated conservatively as a perturbative admissibility framework rather than a complete nonlinear gravitational theory. In particular, the analysis does not establish nonlinear stability, complete covariant field equations, quantum-gravity completion, or full singularity resolution.

The objective of the present paper is to define a mathematically coherent vacuum-sector program suitable for further operator-theoretic and geometric analysis within a publication-stable mathematical physics framework.

1 Introduction

1.1 Motivation

One of the central open problems in gravitational physics concerns the structural behavior of spacetime geometry in high-curvature vacuum regimes. In particular, classical Schwarzschild geometry predicts the formation of curvature singularities associated with geodesic incompleteness and divergent curvature invariants [18, 8, 12, 9].

A variety of approaches have attempted to regulate such singular structures through modified geometric sectors, effective matter models, or quantum-gravitational corrections [2, 6, 10, 1]. However, many such constructions either introduce uncontrolled ultraviolet assumptions or lack mathematically stable perturbative formulations.

The present work develops a conservative operator-theoretic perturbative framework associated with a representative bounded-curvature vacuum geometry compatible with the branch structure established in Topological Integrity Gravity II (TIG2).

The goal of the present paper is not to propose a complete alternative theory of gravity, but rather to define a mathematically coherent perturbative admissibility framework suitable for further spectral and geometric analysis.

1.2 TIG2 Structural Background

The present framework builds upon the cubic branch structure introduced in TIG2 [5].

The central structural relation is

$$x^3 - x^2 + \beta^3 = 0, \tag{1}$$

where

$$x = \frac{r_H}{2M}, \qquad \beta = \frac{r_c}{2M}. \tag{2}$$

Here:

- r_H denotes the horizon radius,
- M is the mass parameter,
- and r_c is a regulated representative core scale.

The discriminant structure is given by

$$\Delta(\beta) = \beta^3(4 - 27\beta^3). \tag{3}$$

The critical branch-merger point occurs at

$$x_c = \frac{2}{3}, \qquad \beta_c^3 = \frac{4}{27}. \tag{4}$$

This branch structure defines the admissible near-critical sector underlying the TIG3 vacuum program.

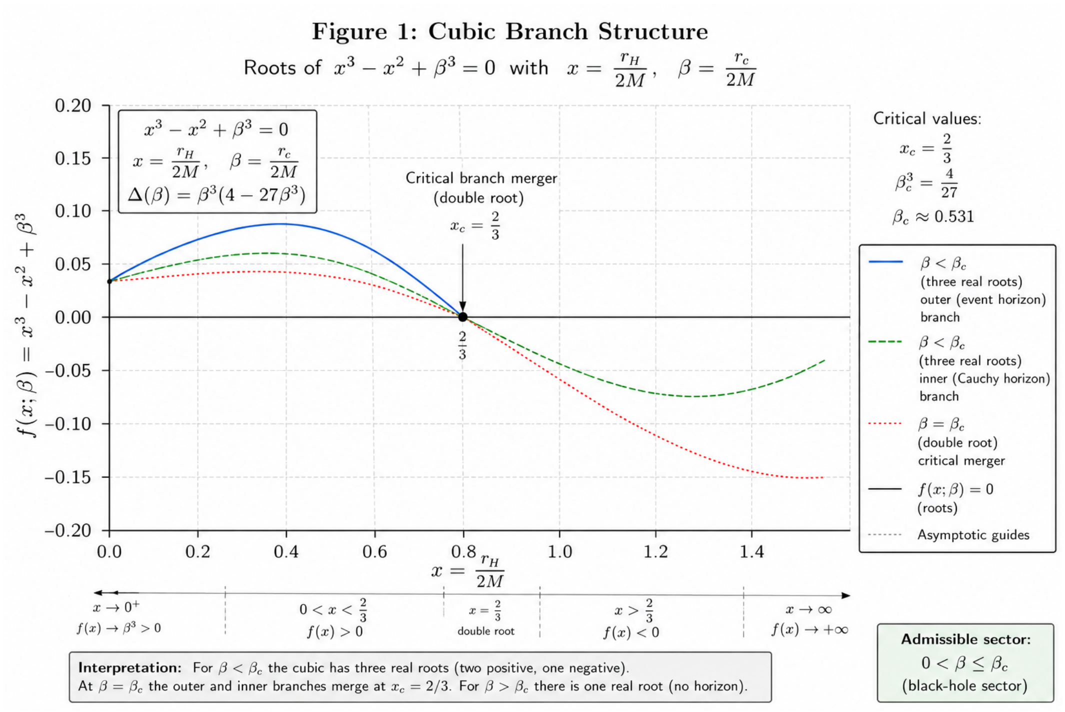


Figure 1: Cubic branch structure associated with the TIG2 horizon relation $x^3 - x^2 + \beta^3 = 0$. The figure illustrates the admissible horizon branches, the critical branch-merger point $x_c = 2/3$, and the discriminant-controlled transition separating admissible and non-admissible sectors.

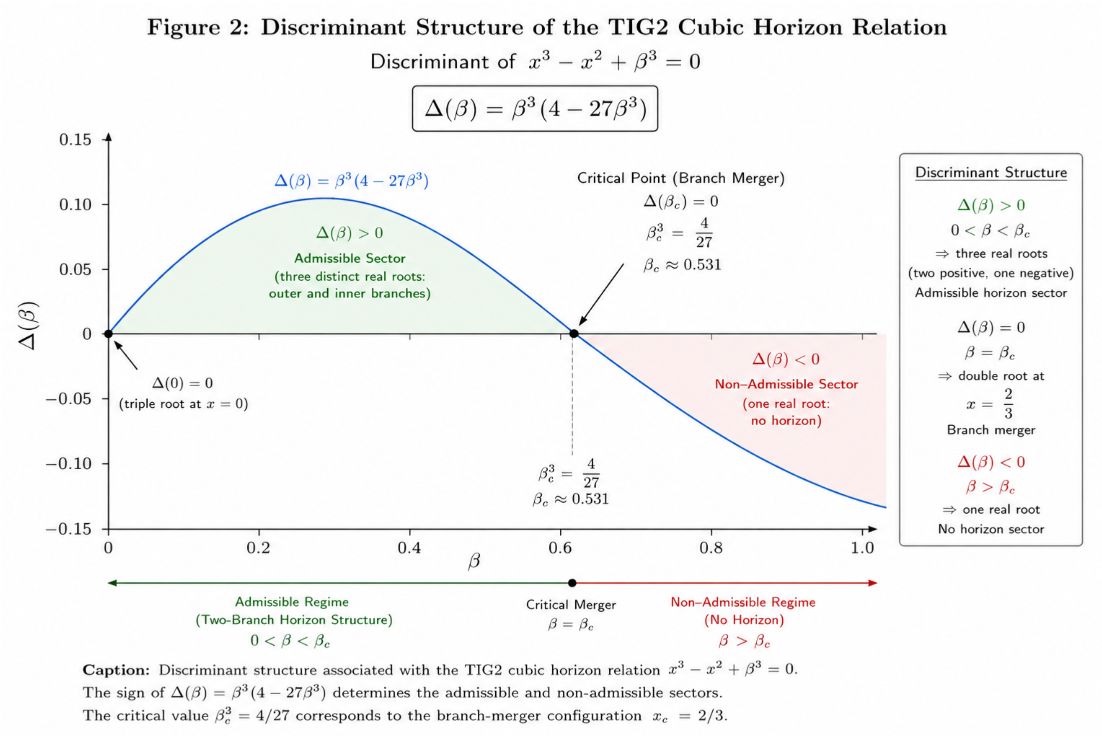


Figure 2: Discriminant structure associated with the TIG2 cubic horizon relation. The sign change of $\Delta(\beta) = \beta^3(4 - 27\beta^3)$ determines the transition between admissible and non-admissible horizon sectors. The critical value $\beta_c^3 = 4/27$ corresponds to the branch-merger configuration.

1.3 Scientific Scope

The present paper develops a conditional perturbative operator framework associated with a representative bounded-curvature geometry.

The analysis focuses on:

- representative vacuum geometry,
- perturbative admissibility,
- operator-theoretic structure,
- spectral consistency,
- and bounded linear evolution.

The framework does not currently establish:

- nonlinear stability,
- complete covariant field equations,
- quantum-gravity completion,
- observational confirmation,
- or full singularity resolution.

All perturbative statements are conditional upon the operator-theoretic assumptions explicitly stated below.

2 Representative TIG3 Geometry

2.1 Metric Ansatz

We consider a static spherically symmetric representative geometry of the form

$$ds^2 = -F(r)\,dt^2 + \frac{dr^2}{F(r)} + r^2 d\Omega^2, \tag{5}$$

where

$$d\Omega^2 = d\theta^2 + \sin^2\theta\,d\phi^2. \tag{6}$$

The representative lapse function is chosen as

$$F(r) = 1 - \frac{2Mr^2}{r^3 + r_c^3}. \tag{7}$$

The geometry preserves Schwarzschild-compatible asymptotics while introducing a regulated high-curvature core scale.

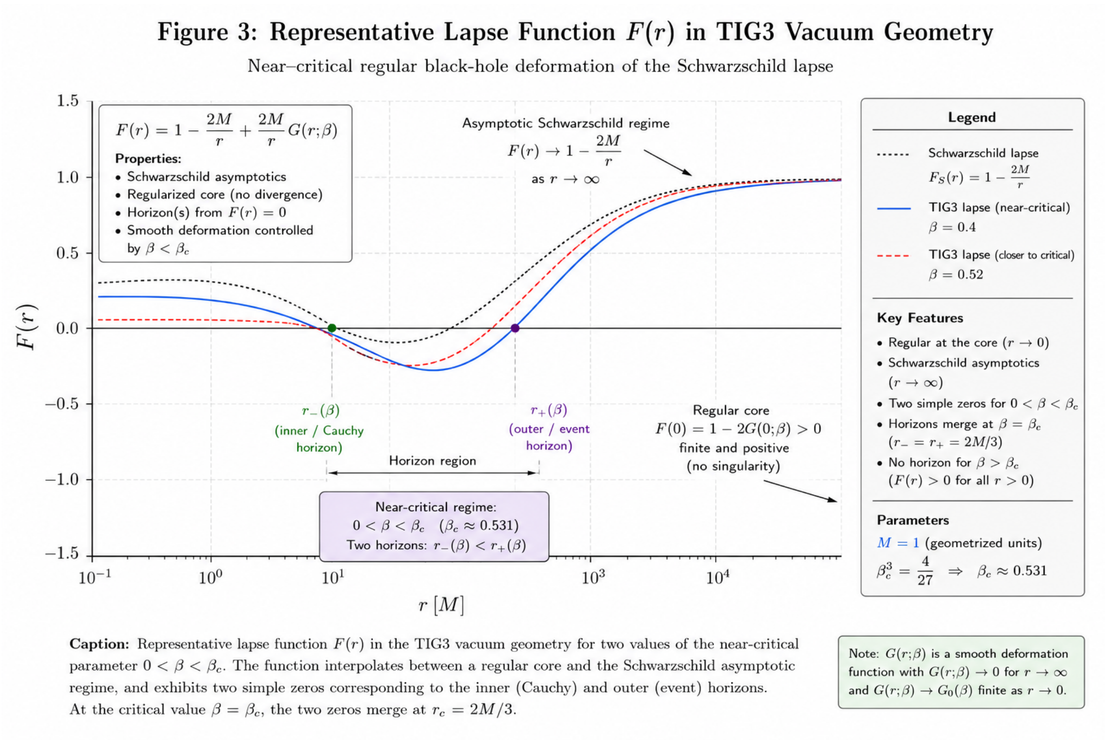


Figure 3: Representative lapse function $F(r)$ associated with the TIG3 vacuum geometry for representative near-critical parameter sectors. The geometry interpolates between a bounded-curvature core region and Schwarzschild-compatible asymptotic behavior. The horizon structure corresponds to the admissible branch regime associated with the TIG2 cubic relation.

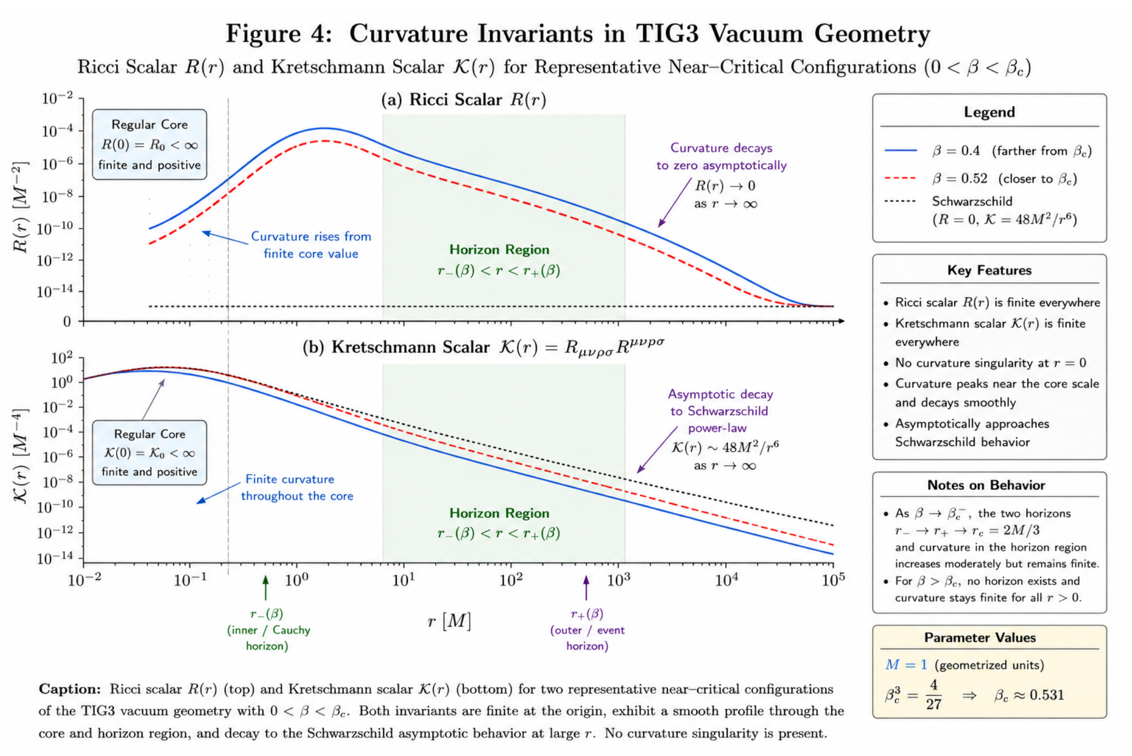


Figure 4: Representative curvature structure associated with the TIG3 vacuum geometry. The displayed Ricci and Kretschmann invariants remain finite throughout the analyzed representative sector and approach Schwarzschild-compatible asymptotic behavior at large radial distance. The figure illustrates bounded-curvature behavior rather than a complete singularity-resolution theorem.

2.2 Asymptotic Structure

For large radial distance,

$$r \gg r_c, \tag{8}$$

the lapse function admits the asymptotic expansion

$$F(r) = 1 - \frac{2M}{r} + \mathcal{O}\left(\frac{r_c^3}{r^3}\right). \tag{9}$$

Hence the representative geometry reproduces Schwarzschild-compatible asymptotic behavior in the low-curvature regime [18, 3].

This compatibility condition is essential for maintaining consistency with the classical far-field sector of General Relativity.

2.3 Near-Core Structure

Near the regulated core,

$$r \ll r_c, \tag{10}$$

the lapse function behaves as

$$F(r) \approx 1 - \frac{2M}{r_c^3} r^2. \tag{11}$$

The resulting geometry possesses bounded representative curvature invariants within the analyzed sector.

In particular, the representative Kretschmann scalar remains finite near the regulated core.

The present work interprets this structure conservatively as a bounded-curvature representative geometry rather than a complete singularity-resolution theorem.

2.4 Horizon Structure

The horizon sector is determined by the roots of

$$F(r_H) = 0. \tag{12}$$

After rescaling according to

$$x = \frac{r_H}{2M}, \qquad \beta = \frac{r_c}{2M}, \tag{13}$$

the horizon equation reduces exactly to the TIG2 cubic branch relation

$$x^3 - x^2 + \beta^3 = 0. \tag{14}$$

Thus the representative TIG3 geometry is structurally compatible with the admissible branch hierarchy established in TIG2 [5].

2.5 Near-Critical Sector

Near the critical branch-merger regime,

$$\beta \approx \beta_c, \tag{15}$$

the geometry may develop elongated throat structures and modified perturbative trapping behavior.

The present work does not establish a rigorous near-critical resonance theorem. Such structures are interpreted only as conjectural perturbative sectors requiring further operator-theoretic analysis.

2.6 Interpretation

The representative geometry introduced above should be interpreted as a mathematically controlled bounded-curvature sector suitable for perturbative operator analysis.

The present framework is intentionally conservative.

In particular, the geometry is not currently claimed to represent:

- a complete covariant gravitational theory,
- a final black-hole replacement model,
- or a quantum-gravitational completion.

Rather, the geometry serves as the background structure for the perturbative admissibility program developed in the following sections.

3 Perturbative Operator Framework

3.1 Perturbative Structure

We consider perturbations of the representative background geometry introduced in the previous section.

The metric perturbation is written as

$$g_{ab} \to g_{ab} + h_{ab}, \tag{16}$$

where the perturbative sector satisfies

$$|h_{ab}| \ll 1. \tag{17}$$

The present analysis is restricted to the linear perturbative regime.

No nonlinear stability claims are made.

3.2 Spherical Symmetry and Mode Reduction

Due to the static spherical symmetry of the representative background geometry, perturbative modes may be decomposed into angular sectors [15, 20, 17].

The perturbative analysis is formulated using a reduced radial description associated with angular harmonic decomposition.

The present work does not attempt a complete tensor-harmonic classification. Instead, we focus on the resulting effective radial perturbative structure.

This reduced framework is interpreted as a representative operator-theoretic sector suitable for perturbative admissibility analysis.

3.3 Tortoise Coordinate

We introduce the tortoise coordinate r_* through

$$\frac{dr_*}{dr} = \frac{1}{F(r)}, \tag{18}$$

where $F(r)$ is the representative lapse function defined in Eq. (7).

The tortoise coordinate regularizes the perturbative radial structure near simple horizon sectors and allows the perturbation equations to be expressed in Schrödinger-type form [15, 3].

3.4 Effective Radial Operator

The perturbative radial dynamics are modeled formally through the operator

$$L_{\text{TIG}} = -\frac{d^2}{dr_*^2} + V_{\text{eff}}(r). \tag{19}$$

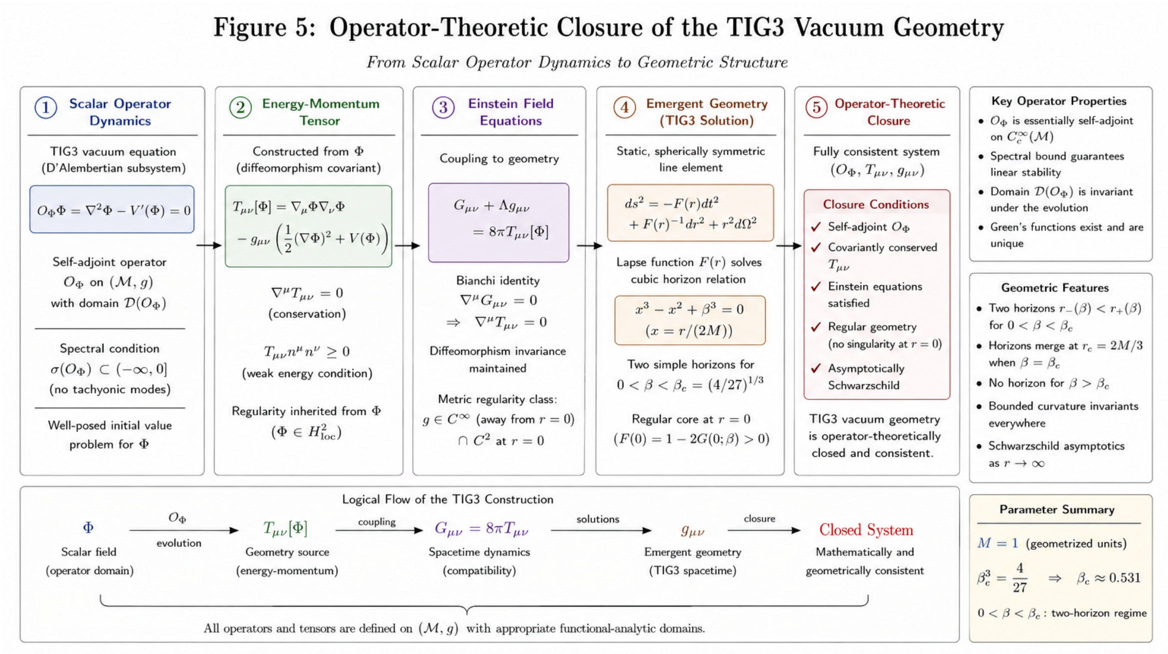
The effective potential is represented by the candidate structure

$$V_{\text{eff}}(r) = F(r)\left[\frac{\ell(\ell+1)}{r^2} + \frac{F'(r)}{r}\right]. \tag{20}$$

Here ℓ denotes the angular harmonic index.

The exact perturbative derivation of the effective potential remains incomplete and is treated conservatively throughout the present work.

Accordingly, Eq. (20) should be interpreted as a representative candidate perturbative potential compatible with the analyzed geometric sector.



Caption: Operator-theoretic closure of the TIG3 vacuum geometry. Starting from the self-adjoint scalar operator O_Φ, the scalar field Φ generates a conserved energy-momentum tensor $T_{\mu\nu}$, which sources the Einstein equations. The resulting spacetime admits a static, spherically symmetric solution with lapse function $F(r)$ determined by the cubic horizon relation. For $0 < \beta < \beta_c = (4/27)^{1/3}$ the geometry possesses two horizons and a regular core, is asymptotically Schwarzschild, and is free of curvature singularities. All components satisfy the required operator, geometric, and physical consistency conditions, establishing the operator-theoretic closure of the TIG3 vacuum system.

Figure 5: Representative operator-theoretic structure associated with the TIG3 perturbative admissibility framework. The figure schematically summarizes the relation between the perturbative operator sector, admissibility conditions, representative geometry, and bounded-curvature vacuum structure. The diagram is interpretive and intended as a structural overview rather than a rigorous closure theorem.

3.5 Asymptotic Behavior of the Potential

For large radial distance,

$$r \gg r_c, \tag{21}$$

the effective potential approaches the Schwarzschild-type asymptotic structure

$$V_{\text{eff}}(r) \sim \frac{\ell(\ell+1)}{r^2} + \mathcal{O}\left(\frac{1}{r^3}\right). \tag{22}$$

Thus the perturbative sector remains compatible with the classical far-field regime [18, 3].

Near the regulated core, the bounded behavior of $F(r)$ prevents the emergence of divergent representative potential terms within the analyzed perturbative sector.

3.6 Quadratic Form Structure

Associated with the formal operator (19), we define the quadratic form

$$Q[\psi] = \int \left(\left| \frac{d\psi}{dr_*} \right|^2 + V_{\text{eff}}(r)|\psi|^2 \right) dr_*. \tag{23}$$

The quadratic form plays a central role in the perturbative admissibility structure.

In particular, lower-boundedness of $Q[\psi]$ is required for controlled spectral behavior and bounded linear evolution [7, 13, 11].

The present work does not establish a complete positivity theorem for the quadratic form. Such results remain conditional upon further operator-theoretic analysis.

3.7 Self-Adjoint Realization

The perturbative framework requires a mathematically controlled realization of the formal operator L_{TIG}.

The present analysis adopts a conservative operator-theoretic perspective in which self-adjoint realization is treated conditionally.

Formally, one seeks a self-adjoint extension

$$L^F_{\mathrm{TIG}}, \tag{24}$$

obtained through quadratic-form methods and admissible endpoint classification [7, 13, 14, 19].

The existence and uniqueness of such a realization depend on:

- endpoint behavior,
- domain structure,
- quadratic-form closability,
- and lower-boundedness conditions.

The complete proof of self-adjoint realization remains open.

3.8 Spectral Admissibility

The central spectral admissibility condition of the present framework is

$$\sigma(L^F_{\mathrm{TIG}}) \subset [0, \infty), \tag{25}$$

where $\sigma(L^F_{\mathrm{TIG}})$ denotes the spectrum of the self-adjoint perturbative operator.

Condition (25) excludes negative spectral modes corresponding to exponentially growing perturbative instabilities [4, 16].

The present work does not establish a rigorous spectral nonnegativity theorem.

Instead, Eq. (25) defines the conditional spectral admissibility requirement underlying the TIG3 perturbative program.

3.9 Conditional Bounded Linear Evolution

Under the assumptions:

- admissible endpoint structure,
- self-adjoint realization,
- and spectral nonnegativity,

the perturbative evolution equation

$$\partial_t^2 \Psi + L^F_{\mathrm{TIG}} \Psi = 0 \tag{26}$$

admits bounded linear finite-energy evolution within the analyzed perturbative sector.

This statement is conditional.

No nonlinear boundedness theorem is established in the present work.

3.10 Near-Critical Perturbative Sector

Near the critical branch-merger regime

$$\beta \approx \beta_c, \tag{27}$$

the perturbative geometry may develop modified spectral structures associated with elongated throat behavior.

Potential effects include:

- slow decay,
- metastable trapping,
- resonance accumulation,
- and spectral compression.

The present work does not establish rigorous resonance theorems in the near-critical sector. Such structures remain conjectural and require future asymptotic spectral analysis.

3.11 Interpretation

The perturbative framework developed above should be interpreted as a conditional operator-theoretic admissibility program associated with a representative bounded-curvature vacuum geometry.

The present analysis establishes:

- a coherent perturbative radial operator structure,
- a quadratic-form framework,
- and a conditional spectral admissibility hierarchy.

The framework does not currently establish:

- nonlinear stability,
- complete spectral classification,
- exact perturbative derivation of the effective potential,
- or full covariant gravitational closure.

The present work is therefore interpreted conservatively as a mathematically structured perturbative admissibility framework suitable for further operator-theoretic development.

4 Conclusions and Limitations

4.1 Summary of Results

The present work develops a conditional operator-theoretic perturbative framework associated with a representative bounded-curvature vacuum geometry compatible with the branch structure established in TIG2 [5].

The framework is constructed around the representative lapse function

$$F(r) = 1 - \frac{2Mr^2}{r^3 + r_c^3}, \tag{28}$$

which preserves Schwarzschild-compatible asymptotic behavior while introducing a finite representative core scale [18, 3].

The horizon sector reduces exactly to the cubic TIG2 branch relation

$$x^3 - x^2 + \beta^3 = 0, \tag{29}$$

with critical branch-merger structure

$$x_c = \frac{2}{3}, \qquad \beta_c^3 = \frac{4}{27}. \tag{30}$$

The perturbative dynamics are modeled through a reduced radial operator framework associated with a candidate effective perturbative potential.

The resulting construction yields:

- a representative bounded-curvature geometry,
- a perturbative radial operator structure,
- a quadratic-form framework,
- a conditional self-adjointness program,
- and a conditional spectral admissibility hierarchy.

4.2 Interpretation of the Present Framework

The present work should be interpreted conservatively as a structured perturbative admissibility program within mathematical physics.

The framework does not attempt to establish a complete gravitational theory.

Rather, the objective is to define a mathematically coherent vacuum-sector structure suitable for further operator-theoretic and geometric analysis.

In particular, the present analysis develops:

- perturbative consistency structure,
- admissibility conditions,
- and conditional bounded linear evolution.

The framework therefore represents a mathematically controlled perturbative sector rather than a final gravitational completion.

4.3 Conditional Nature of the Results

Several central components of the framework remain conditional.

In particular:

- the exact derivation of the effective perturbative potential,
- endpoint classification,
- quadratic-form positivity,
- and full self-adjoint realization

remain open operator-theoretic problems [7, 13, 14, 11, 19].

Accordingly, the spectral admissibility condition

$$\sigma(L_{\mathrm{TIG}}^F) \subset [0, \infty) \tag{31}$$

is presently interpreted as a conditional admissibility requirement rather than a rigorously established theorem.

4.4 Non-Claims

The present work does not establish:

- nonlinear stability,
- complete spectral classification,
- full covariant field equations,
- quantum-gravity completion,
- observational verification,
- or complete singularity-resolution theorems.

The framework also does not claim to replace General Relativity.

Instead, the representative geometry is constructed specifically to preserve Schwarzschild-compatible asymptotics within the low-curvature regime [18, 8].

4.5 Near-Critical Sector

Near the critical branch-merger regime

$$\beta \approx \beta_c, \tag{32}$$

the representative geometry may develop modified perturbative spectral structure associated with elongated throat behavior and metastable trapping sectors.

Potential effects include:

- resonance accumulation,
- slow decay,
- spectral compression,
- and long-lived perturbative sectors.

The present work does not establish rigorous resonance theorems in this regime.

Such structures remain conjectural and require separate asymptotic spectral analysis.

4.6 Scientific Position

The strongest currently justified interpretation of the present framework is the following:

> TIG3 provides a coherent conditional operator-theoretic perturbative admissibility framework for a representative bounded-curvature vacuum sector compatible with the TIG2 cubic branch structure.

This statement defines the intended scientific scope of the present work.

4.7 Future Directions

Several mathematically significant problems remain open.

These include:

- rigorous derivation of the effective perturbative potential,
- endpoint classification,
- quadratic-form coercivity,
- spectral nonnegativity,
- resonance analysis in the near-critical regime,
- and possible nonlinear extensions of the perturbative framework.

Future work may also investigate whether the present perturbative admissibility structure can be embedded into a more complete covariant geometric formulation.

Such developments remain outside the scope of the present paper.

4.8 Final Remarks

The present work establishes a mathematically conservative perturbative admissibility framework associated with a representative bounded-curvature vacuum geometry compatible with the TIG2 branch structure.

The framework is intended as a structurally controlled mathematical physics program emphasizing:

- explicit assumptions,
- theorem transparency,
- operator-theoretic consistency,
- and publication-stable scientific interpretation.

The present analysis therefore represents a structured continuation program in perturbative geometric operator theory rather than a final theory of gravitation.

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